

Top Business Objectives:

What are the customer's top business objectives? Panasonic Avionics Corporation (PAC) is executing a strategic pivot from a traditional aerospace hardware manufacturer to a cloud-native, software-and-services (SaaS/PaaS) provider 1, 2. Their primary objectives are to improve Return on Equity (ROE) and profitability through high-margin digital services, increase Passenger Engagement Minutes (PEMs) to drive airline ancillary revenue, and reduce "Aircraft on Ground" (AOG) events 3. Environmentally, PAC aims to achieve net-zero CO2 emissions by 2030 in alignment with the "Panasonic GREEN IMPACT" initiative 3.

How are these objectives supported by the customer's initiatives? These objectives are supported by three core initiatives:

1. **Astrova Platform Expansion:** Rolling out their modular, 4K OLED in-flight engagement platform, which enables dynamic digital ecosystems and has already secured over 100 airline programs 1, 4.
2. **Multi-Orbit & LEO Connectivity:** Modernizing in-flight Wi-Fi by adopting Low Earth Orbit (LEO) satellites (via partnerships like Spacesail) to offer streaming-grade "Connectivity-as-a-Service" 1, 2.
3. **The "Panasonic Go" AI Initiative:** Transitioning to an AI-based business model to drive supply chain optimization and localized passenger personalization 1, 5.

How are these initiatives supported by IT, and how are they funded? IT supports these initiatives by aggressively migrating legacy monolithic applications to AWS and Azure cloud architectures, creating an "Avionics Specific Data Lake" to process telemetry and passenger data in near real-time 2, 6. IT is also driving operational efficiency through Event-Driven automation 7. These modernization efforts are funded primarily by eliminating technical debt and escaping ballooning legacy virtualization costs (e.g., VMware) 1, 8.

What is the funding for the initiatives, and what is the decision-making process for determining which initiatives will be funded? Funding is guided by parent company Panasonic Holdings' "selection and concentration" strategy, which leverages strategic investment allocations for high-growth areas like the Avionics "Solutions Domain" 9, 10. The decision-making process is highly disciplined, prioritizing businesses and initiatives where the Return on Invested Capital (ROIC) exceeds the Weighted Average Cost of Capital (WACC) 11. PAC's leadership influence is strong here, as PAC CEO Ken Sain is slated to become Group CEO of Panasonic Holdings in April 2026, cementing Avionics as the template for broader corporate growth 9.

What is the customer's personnel situation, and how are they ensuring they have the necessary resources to support the IT initiatives? PAC is navigating an industry-wide, permanent shortage of certified aviation technicians, alongside risks of institutional knowledge loss from global corporate workforce reductions 12, 13. To secure the necessary resources, PAC is aggressively hiring Cloud, Linux, and Data Reliability Engineers 14. Furthermore, they have opened new Software Centers of Excellence in Portland, Oregon (focused on Linux and Android) and Pune, India, specifically to internalize software expertise and build the next generation of localized digital experiences 15, 16.

Top Business Challenges:

What are the top challenges the customer faces over the next three years (according to their own perspective)?

1. **Unprecedented MRO Demand & Technician Shortages:** Severe supply chain constraints and a 17,000+ commercial aircraft order backlog force airlines to operate older fleets, driving massive Maintenance, Repair, and Overhaul (MRO) demand amidst a global shortage of certified technicians 12.
2. **Rising Technical Debt & Licensing Costs:** The urgent need to migrate off legacy monolithic systems and escape surging VMware hypervisor costs post-Broadcom acquisition 1, 8.
3. **Strict Regulatory Compliance:** Meeting the rigorous EASA Part-IS cybersecurity mandate, which requires an operational Information Security Management System (ISMS) within 18 months of the 2026 deadline 17, 18.
4. **Fierce Competition:** Defending market share from agile competitors like Safran Passenger Innovations (SPI) and Thales, who are rapidly capturing new aircraft delivery programs 19, 20.

What are the potential impacts for the customer if these challenges remain unresolved?

Failure to comply with EASA Part-IS could result in severe financial penalties and the grounding of airline customers' aircraft 18, 21. Unresolved technical debt and reliance on manual IT processes will exacerbate the technician shortage, leading to increased aircraft downtime (AOG) and stalled digital innovation 7, 12. Ultimately, moving too slowly will cause irreversible market share erosion to competitors 19, 20.

How are these challenges being addressed by IT initiatives, and how are those initiatives funded?

IT is addressing these challenges by establishing a unified, automated hybrid-cloud and edge computing architecture, directly funded by cost-savings from legacy IT consolidation 1, 8. This digital pivot connects directly to several strategic Red Hat opportunities:

- To bypass the aviation technician shortage and reduce aircraft downtime, IT is deploying **Event-Driven Ansible (EDA)** integrated with ServiceNow and NetBox to automate MRO tasks and configuration changes 7.
- To escape ballooning VMware costs and securely containerize aging applications, PAC is leveraging **Red Hat OpenShift Virtualization** for a seamless VM-to-Container lift-and-shift 8, 22.
- To meet strict EASA compliance and eliminate deployment downtime on disconnected aircraft, IT is transitioning to immutable edge deployments using **RHEL 10 Image Mode** and **Red Hat Device Edge**, enabling instant rollbacks and FIPS-compliant quantum cryptography 23-25.
- To close internal skill gaps, IT is actively evaluating AI-assisted administration via **RHEL 10 Command Line Assist** and **Red Hat Lightspeed** 26.